EXPLICIT DESCRIPTION OF THE FUNCTORS OF A SECOND LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The second localization of a site. Same chapter, Lemma lemma-relocalize, recorded as the context of this card. Supplies the diagram, the functor j and the identities satisfied by the composites.
- The equivalence between localized sheaves and slices. Same chapter, Lemma lemma-essential-image-j-shriek. Identifies the localized categories of sheaves with the slices, in which the descriptions are stated.

Source reference. Original lemma 00Y1: sources/stacks-sites/pinned-original-source.tex, source lines 5613–5613. Navigation pointer invoked at proof line 5613; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $f : V \to U$ be a morphism of $\mathcal{C}$. Then there exists a commutative diagram*

$$\begin{array}{ccc} \mathcal{C}/V & \xrightarrow{\quad j \quad} & \mathcal{C}/U \\ & {\scriptstyle j_V} \searrow \quad \swarrow {\scriptstyle j_U} & \\ & \mathcal{C} & \end{array}$$

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of continuous and cocontinuous functors. The functor $j : \mathcal{C}/V \to \mathcal{C}/U$, $(a : W \to V) \mapsto (f \circ a : W \to U)$ *is identified with the functor* $j_{V/U} : (\mathcal{C}/U)/(V/U) \to \mathcal{C}/U$ *via the identification* $(\mathcal{C}/U)/(V/U) = \mathcal{C}/V$. *Moreover we have* $j_{V!} = j_{U!} \circ j_!$, $j_V^{-1} = j^{-1} \circ j_U^{-1}$, *and* $j_{V*} = j_{U*} \circ j_*$.

Lemma 0.2. *Notation* $\mathcal{C}$, $f : V \to U$, j_U, j_V, *and* j *as in Lemma 0.1. Via the identifications* $Sh(\mathcal{C}/V) = Sh(\mathcal{C})/h_V^\#$ *and* $Sh(\mathcal{C}/U) = Sh(\mathcal{C})/h_U^\#$ *of Lemma 00Y1 we have*

(1) *the functor* j^{-1} *has the following description*

$$j^{-1}(\mathcal{H} \xrightarrow{\varphi} h_U^\#) = (\mathcal{H} \times_{\varphi, h_U^\#, f} h_V^\# \to h_V^\#).$$

(2) *the functor* $j_!$ *has the following description*

$$j_!(\mathcal{H} \xrightarrow{\varphi} h_V^\#) = (\mathcal{H} \xrightarrow{h_f \circ \varphi} h_U^\#)$$

Proof. Proof of (2). Recall that the identification $Sh(\mathcal{C}/V) \to Sh(\mathcal{C})/h_V^\#$ sends $\mathcal{G}$ to $j_{V!}\mathcal{G} \to j_{V!}(*) = h_V^\#$ and similarly for $Sh(\mathcal{C}/U) \to Sh(\mathcal{C})/h_U^\#$. Thus $j_!\mathcal{G}$ is mapped to $j_{U!}(j_!\mathcal{G}) \to j_{U!}(*) = h_U^\#$ and (2) follows because $j_{U!}j_! = j_{V!}$ by Lemma 0.1.

The reader can now prove (1) by using that j^{-1} is the right adjoint to $j_!$ and using that the rule in (1) is the right adjoint to the rule in (2). Here is a direct proof. Suppose that $\varphi : \mathcal{H} \to h_U^\#$ is an object of $Sh(\mathcal{C})/h_U^\#$. By the proof of Lemma 00Y1 this corresponds to the sheaf $\mathcal{H}_\varphi$ on $\mathcal{C}/U$ defined by the rule

$$(a : W \to U) \longmapsto \{s \in \mathcal{H}(W) \mid \varphi(s) = a\}$$

on $\mathcal{C}/U$. The pullback $j^{-1}\mathcal{H}_\varphi$ to $\mathcal{C}/V$ is given by the rule

$$(a : W \to V) \longmapsto \{s \in \mathcal{H}(W) \mid \varphi(s) = f \circ a\}$$

by the description of $j^{-1} = j_{U/V}^{-1}$ as the restriction of $\mathcal{H}_\varphi$ to $\mathcal{C}/V$. On the other hand, applying the rule to the object

$$\mathcal{H}' = \mathcal{H} \times_{\varphi, h_U^\#, f} h_V^\# \xrightarrow{\varphi'} h_V^\#$$

of $Sh(\mathcal{C})/h_V^\#$ we get $\mathcal{H}'_{\varphi'}$ given by

$$\begin{aligned}(a : W \to V) \longmapsto & \{s' \in \mathcal{H}'(W) \mid \varphi'(s') = a\} \\ = & \{(s, a') \in \mathcal{H}(W) \times h_V^\#(W) \mid a' = a \text{ and } \varphi(s) = f \circ a'\}\end{aligned}$$

which is exactly the same rule as the one describing $j^{-1}\mathcal{H}_\varphi$ above. □